MILITARY SPECIFICATION

SQUIB SWITCHES M12, M31, M32, M33, M34, M35, M36

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers requirements, quality assurance provisions, and the preparation for delivery criteria for two type of squib switches which are used in the Sergeant warhead section.

1.2 Classification. This specification covers Squib Switches of the following types:

Type I - Delay switch entitled Squib Switches: M12 and M31.

Type II - Delay squib activated switches entitled Squib Switches: M32, M33, M34, M35 and M36.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-E-5272	- Environmental Testing, Aeronautical and Associated Equipment, General Specification For.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-271	- Nondestructive Testing and Definitions.
MIL-STD-810	- Environmental Test Methods.

FEDERAL

FED-STD-356	- Commercial Packaging of Supplies and Equipment.

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DRAWINGS

8829810	- Squib Switch, Delay M31.
8829814	- Squib Switch, Delay M12.
8830148	- Squib Switch M33.
8830149	- Squib Switch M35.
8830150	- Squib Switch M32.
8830151	- Squib Switch M34.
8838421	- Housing.
8840149	- Squib Switch M36.
8844571	- Housing.
EL-8867522	- Equipment List.
EL-8871618	- Equipment List.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The switch shall comply with all requirements specified on the drawing (dwg) 8829810, 8829814, 8830148, 8830149, 8830150, 8830151 or 8840149 as applicable, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.4 Serialization. The contractor shall assign serial numbers as required by drawings. The contractor may select any number which complies with the drawing requirement. These numbers shall not be duplicated.

3.5 Squib energy.

3.5.1 Non-function.

3.5.1.1 Type I. The switch shall not function when the squib is subjected to 15,000 ergs. Furthermore, the switch shall not function when the squib is subjected to 100 milliampere (ma) for a minimum of 10 seconds. Testing shall be as specified in 4.4.1.

3.5.1.2 Type II. The switch shall not function when the squib is subjected to 15,000 ergs. Furthermore, the switch shall not function when the squib is subjected to 100 milliampere (ma) for a minimum of 10 seconds. Testing shall be as specified in 4.4.1.

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3.5.2 Functioning.

3.5.2.1 Type I. The switch shall function within 1 plus or minus (+) 0.5 seconds (sec) for the M31 and 0.1 ± 0.05 sec, for the M12 after the squib is subjected to 42,000 ergs of energy when tested as specified in 4.4.2.

3.5.2.2 Type II. The switch shall function within 0.012 seconds after the squib is subjected to 42,000 ergs of energy when tested as specified in 4.4.2.

3.6 Contact position and resistance.

3.6.1 Prior to functioning.

3.6.1.1 Type I. The position of the contacts shall be as specified in Table I and the closed contact resistance shall be 0.008 ohm maximum when tested as specified in 4.4.3.

Table I

Switch Drawing	Normally Closed Contacts	Normally Open Contacts
8829810	1 and 2	3 and 4
8829814	1, 2 and 3	4
8840149	1 and 2	3, 4, 5 and 6
8830149	1	2, 3 and 4
8830148	1, 2 and 3	4
8830151	1 and 2	3 and 4
8830150	1, 2, 3 and 4	none

3.6.1.2 Type II. The provision of 3.6.1.1 apply to Type II switches.

3.6.2 After functioning.

3.6.2.1 Type I. The position of the contacts shall be in the reverse of the prior to functioning position and the closed contact resistance shall be 0.008 ohm maximum for Type I when tested as specified in 4.4.3.

3.6.2.2 Type II. The position of the contacts shall be in the reverse of the prior to functioning position and the closed contact resistance shall be 0.055 ohm maximum when tested as specified in 4.4.3.

3.7 Bridge resistance.

3.7.1 Type I switches. The bridge resistance shall be 2.2 + 0.3 ohm for the Type I squib switches when tested as specified in 4.4.4.

3.7.2 Type II switches. The bridge resistance shall be 1.8 + 0.2 ohm for the Type II squib switches when tested as specified in 4.4.4.

3.8 Insulation resistance.

3.8.1 Prior to functioning.

3.8.1.1 Type I. The insulation resistance of the switch shall not be less than 50 megohms between the points listed in Table II when tested as specified in 4.4.5.

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Table II

Insulation resistance before functioning

Between	And
Both pins of 1	Housing plus both pins of 2, 3 and 4
Both pins of 2	Housing plus both pins of 3 and 4
Both pins of 3	Housing plus both pins of 4
Both pins of 4	Housing
Shunted squib leads	Housing plus pins 1 and 2. Both pins of 4 and pin 3 (or both pins of 3 when N.O.)
One pin of 3	Other pin of 3 (when open)
One pin of 4	Other pin of 4

3.8.1.2 Type II. The insulation resistance between each test point and all other test points listed in Table III shall be greater than 50 megohms when tested as specified in 4.4.5.

Table III

1. Case
2. Shunted squib leads
3. Each terminal, except pairs of terminals connected by normally closed contacts shall be considered as single test points.

3.8.2 After functioning.

3.8.2.1 Type I. The insulation resistance of the switch after functioning shall not be less than 50 megohms between the points listed in Table IV when tested as specified in 4.4.5.

Table IV

Between	And
Both pins of 1	Housing plus both pins of 2, 3 and 4
Both pins of 2	Housing plus both pins of 3 and 4
Both pins of 3	Housing plus both pins of 4
Both pins of 4	Housing
One pin of 1	Other pin of 1
One pin of 2	Other pin of 2 (when open)
One pin of 3	Other pin of 3 (when open)

3.8.2.2 Type II. The insulation resistance of the switch after functioning shall be greater than 20 megohms, when measured between each terminal and case (except pairs of terminals connected by normally closed contacts shall be considered as a single test point), when tested as specified in 4.4.5.

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3.9 Environmental testing. The switch shall comply with MIL-STD-810, visual inspection and failure criteria, section 3. The switch shall also comply with applicable sections in 4.3.3.1 before and after each environmental test. Testing shall be as specified in 4.4.6. Tests shall be conducted in the order specified.

3.9.1 Low temperature. The switch shall show no evidence of damage or deterioration upon exposure to -65 ± 5 degrees Fahrenheit (°F) and while at this temperature shall meet the requirements of Squib energy (3.5.1.1 for Type I switches and 3.5.1.2 for Type II switches), contact position (3.6.1.1 for Type I switches and 3.6.1.2 for Type II switches), bridge resistance (3.7.1 for Type I switches and 3.7.2 for Type II switches), insulation resistance (3.8.1.1 for Type I switches and 3.8.1.2 for Type II switches) and squib energy (3.5.2.1 for Type I switches and 3.5.2.2 for Type II switches) when tested as specified in 4.4.6.1.

3.9.2 High temperature. The switch shall show no evidence of damage or deterioration upon exposure to 160 ± 5°F and while at this temperature shall meet the requirements of Squib energy (3.5.1.1 for Type I switches and 3.5.1.2 for Type II switches), contact position (3.6.1.1 for Type I switches and 3.6.1.2 for Type II switches), bridge resistance (3.7.1 for Type I switches and 3.7.2 for Type II switches), insulation resistance (3.8.1.1 for Type I switches and 3.8.1.2 for Type II switches) and squib energy (3.5.2.1 for Type I switches and 3.5.2.2 for Type II switches) when tested as specified in 4.4.6.2.

3.9.3 Shock. The switch shall show no evidence of damage when subjected to shocks of 30 ± 5 g's three times in each direction along the three mutually perpendicular axes for a total of 18 shocks, when tested as specified in 4.4.6.3. Upon completion of these shocks the switch shall meet the requirements of Squib energy (3.5.1.1 for Type I switches and 3.5.1.2 for Type II switches), contact position (3.6.1.1 for Type I switches and 3.6.1.2 for Type II switches), bridge resistance (3.7.1 for Type I switches and 3.7.2 for Type II switches), insulation resistance (3.8.1.1 for Type I switches and 3.8.1.2 for Type II switches) and squib energy (3.5.2.1 for Type I switches and 3.5.2.2 for Type II switches). After complying with these requirements, the switch shall withstand a repeat of the 18 shocks as specified above and then meet the requirements of contact position and resistance (3.6.2) and insulation resistance (3.8).

3.9.4 Transportation vibration. The switch shall withstand vibration for 15 minutes in each of three mutually perpendicular axes at the frekuencies and amplitudes specified in Table V when tested as specified in 4.4.6.4. The frequency shall be varied uniformly from 8 to 500 to 8 Hertz (Hz) every 15 minutes. Following the vibrations the switch shall meet the requirements of contact position (3.6.1.1 for Type I

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switches and 3.6.1.2 for Type II switches), bridge resistance (3.7.1 for Type I switches and 3.7.2 for Type II switches) and insulation resistance (3.8.1.1 for Type I switches and 3.8.1.2 for Type II switches) followed by operational vibration (3.9.5).

Table V

Frequency Range (Hz)	Cycle Time	Constant Acceleration or Double Amplitude
8-55-8	7-1/2 min	0.060 ± 0.006 inch
55-500-55	7-1/2 min	10 ± 1 gravity (g)

3.9.5 Operational vibration. The switch shall withstand without evidence of damage or functioning a programmed sequence of band-limited white Gaussian noise and combined noise and sine wave vibration when tested as specified in 4.4.6.5. The total time of test shall be 210 seconds: 70 seconds in each of three mutually perpendicular axes. The complex vibration test will consist of the following five sections programmed continuously:

(1) White Gaussian noise at a level of 22 g root mean square (rms) band-limited between 15 and 1500 Hertz (Hz) for one (1) second.
(2) A 4 g rms swept sine wave superimposed on 7.5 g rms of white Gaussian noise band-limited between 15 and 1500 Hz for 15 seconds. The sine wave shall be swept from 800 to 1300 Hz at rate increasing directly with frequency, one such sweep taking the full 15 seconds.
(3) An 8 g rms swept sine wave superimposed on 15 g rms of white Gaussian noise band-limited between 15 and 1500 Hz for 15 seconds. The sine wave shall be swept from 1300 to 800 Hz at a rate decreasing directly with frequency, one such sweep taking the full 15 seconds.
(4) A 4 g sine wave superimposed on 7.5 g rms of white Gaussian noise band-limited between 15 and 1500 Hz for 28 seconds. The sine wave shall be swept from 15 to 1500 Hz and back to 15 Hz at a rate increasing directly with frequency. One such cycle shall take the full 38 seconds.
(5) White Gaussian noise at a level of 22 g rms band-limited between 15 and 1500 Hz for one (1) second.

During section 1 of the final sweep the switch shall meet the requirements of squib energy, functioning (3.5.2.1 for Type I switches and 3.5.2.2 for Type II switches). After completion of the test, the switch shall meet the requirements of contact position and resistance (3.6.2.1 for Type I switches and 3.6.2.2 for Type II switches) and insulation resistance (3.8.2.1 for Type I switches and 3.8.2.2 for Type II switches).

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3.9.6 Leakage (applicable to Type II switches only). The switch shall not leak when immersed in 2 inches of water and the atmosphere above the surface is reduced to 5 ± 1 pounds per square inch (psi) absolute when tested as specified in 4.4.6.6.

3.9.7 Temperature humidity (applicable to Type II only). The switch shall withstand humidity of 95 ± 5 percent at 160 ± 5°F without evidence of damage or affecting functioning when tested as specified in 4.4.6.7. Upon completion of the test, the switch shall meet the requirements of Squib energy (3.5.1.2), contact position (3.6.2.2), bridge resistance (3.7.2), insulation resistance (3.8.2.2) followed by Squib energy (3.5.2.2), contact position (3.6.2.2), bridge resistance (3.7.2) and insulation resistance (3.8.2.2 when tested as specified in 4.4.6.

3.10 X-ray. The squib switch shall show no evidence of loose foreign material, or of missing, misplaced, damaged or incorrect type of contact, slider, bridge, wire or piston, when examined by X-ray as specified in 4.4.7.

3.11 Workmanship.

3.11.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.11.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.11.3 Burrs. Parts shall be free from burrs.

3.11.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.11.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

3.11.6 Electrical. Wires shall be dressed in such a manner as to prevent pinching or chafing of the wire insulation. Ends of wires shall be free from burrs, sharp edges, or points. All parts shall be free from flux, chips, dirt, grease, rust or foreign material.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 First article inspection.

4.2.1 Quantity to be submitted. When the contract requires first article inspection, the contractor shall submit 42 complete assemblies (see dwgs 8829810 and 8829814) for first article testing to the testing facility designated in the invitation for bids or request for proposal.

4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in Table VI for Type I Squib Switches and Table VII for Type II Squib Switches. Tests shall be conducted in the order specified in Table VI and VII.

Table VI

First article inspections and tests

Inspection or Test	No. of Sample Units Tested	No. of Sample Units Non-Function Tested & Then Functioned	Dest	Non-Dest
1. Classification of characteristics (see 4.3.2.1)	42	0		X
2. Contact position and resistance (see 3.6)	42	0		X
3. Bridge resistance (see 3.7)	42	0		X
4. Insulation resistance (see 3.8)	42	0		X
5. X-ray (see 3.10)	42	0		X
6. Squib energy, non-function (see 3.5.1)	42	0		X
7. Squib energy, functioning (see 3.5.2)	7	7	X	
8. Low temperature (see 3.9.1)	35	7	X	
9. High temperature (see 3.9.2)	28	7	X	
10. Shock (see 3.9.3)	21	7	X	
11. Transportation vibration (see 3.9.4)	14	7		X
12. Operational vibration (see 3.9.5)	7	7	X	

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Table VII

First article inspection and tests

Inspection or Test	No. of Sample Units Tested	No. of Sample Units Non-Function Tested & Then Functioned	Dest	Non-Dest
1. Classification of characteristics (see 4.3.2.1)	42	0		X
2. Leakage (see 3.9.6)	42	0		X
3. X-ray (see 3.10)	42	0		X
4. Bridge resistance (see 3.7)	42	0		X
5. Insulation resistance (see 3.8)	42	0		X
6. Contact position and resistance (see 3.6)	42	0		X
7. Squib energy, non-function (see 3.5.1)	42	0		X
8. Squib energy, functioning (see 3.5.2)	6	6	X	
9. Low temperature (see 3.9.1)	36	6	X	
10. High temperature (see 3.9.2)	30	6	X	
11. Shock (see 3.9.3)	24	6	X	
12. Transportation vibration (see 3.9.4)	18	6	X	
13. Operational vibration (see 3.9.5)	12	6	X	
14. Temperature humidity (see 3.9.7)	6	6	X	

4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Quality conformance inspection.

4.3.1 Product inspection examination. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table (see Table VIII and Table IX), and inspected for all characteristics in the applicable classification. Any characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

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Table VIII
Major Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-72	all	140-141	124	256-259	177	541-549	230
73	72	142	125	260-262	178	550-559	231
74	73	143-144	126	263-265	179	560-569	232
75	74	145	127	266-268	180	570-579	233
76	75	146-147	128	269-271	181	580-590	234
77	76	148-149	129	272-275	182	591-601	235
78	77	150	130	276-278	183	602-613	236
79	78	151-152	131	279-282	184	614-625	237
80	79	153-154	132	283-285	185	626-637	238
81-82	80	155-156	133	286-289	186	638-649	239
83	81	157	134	290-293	187	650-662	240
84	82	158-159	135	294-296	188	663-676	241
85	83	160-161	136	297-300	189	677-690	242
86	84	162-163	137	301-304	190	691-704	243
87	85	164-165	138	305-308	191	705-719	244
88-89	86	166-167	139	309-312	192	720-735	245
90	87	168	140	313-316	193	736-751	246
91	88	169-170	141	317-320	194	752-768	247
92	89	171-172	142	321-324	195	769-785	248
93	90	173-174	143	325-329	196	786-804	249
94-95	91	175-176	144	330-333	197	805-823	250
96	92	177-178	145	334-338	198	824-842	251
97	93	179-180	146	339-342	199	843-863	252
98	94	181-182	147	343-347	200	864-884	253
99-100	95	183-184	148	348-352	201	885-907	254
101	96	185-187	149	353-356	202	908-930	255
102	97	188-189	150	357-361	203	931-955	256
103	98	190-191	151	362-366	204	956-981	257
104-105	99	192-193	152	367-372	205	982-1008	258
106	100	194-195	153	373-377	206	1009-1036	259
107	101	196-197	154	378-382	207	1037-1066	260
108-109	102	198-200	155	383-388	208	1067-1098	261
110	103	201-202	156	389-393	209	1099-1131	262
111	104	203-204	157	394-399	210	1132-1166	263
112-113	105	205-207	158	400-405	211	1167-1204	264
114	106	208-209	159	406-411	212	1205-1243	265
115	107	210-212	160	412-417	213	1244-1285	266
116-117	108	213-214	161	418-424	214	1286-1330	267
118	109	215-217	162	425-430	215	1331-1378	268
119	110	218-219	163	431-437	216	1379-1429	269
120-121	111	220-222	164	438-443	217	1430-1483	270
122	112	223-224	165	444-450	218	1484-1542	271
123-124	113	225-227	166	451-457	219	1543-1605	272
125	114	228-230	167	458-465	220	1606-1673	273
126-127	115	231-232	168	466-472	221	1674-1747	274
128	116	233-235	169	473-480	222	1748-1827	275
129-130	117	236-238	170	481-488	223	1828-1914	276
131	118	239-241	171	489-496	224	1915-2010	277
132-133	119	242-243	172	497-504	225	2011-2115	278
134	120	244-246	173	505-513	226	2116-2232	279
135-136	121	247-249	174	514-521	227	2233-2361	280
137	122	250-252	175	522-530	228	over 2361	298
138-139	123	253-255	176	531-540	229		

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Table IX
Minor Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

4.3.2.1 Classification of characteristics (see 3.2). Critical, major and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Squib switch (dwg 8829810 or dwg 8829814) (Type I switches only).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Terminal pin locations	Gage
	102. Case damaged	Visual
	103. Objective evidence relative to material missing or improper (see 3.1)	Visual
	104. Serialization missing or improper (see 3.4)	Visual
Minor:		
	201. Illegible markings on switch	Visual
	202. Evidence of poor workmanship (see 3.11)	Visual

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4.3.2.1.2 Housing (dwg 8838421 and 8844571) (Type II switches only).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Presence of attached and loose metal slivers inside housing (Microscopic examination 20 magnification minimum)	Visual
102.	Objective evidence relative to material missing or improper (see 3.1)	Visual
103.	Evidence of poor workmanship (see 3.11)	Visual
Minor:		
201.	Serialization improper or missing (see 3.4)	Visual

4.3.2.1.3 Squib switches M32, M33, M34, M35 and M36 (dwgs 8830150, 8830148, 8830151, 8830149 and 8840149) (Type II switches only).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Mislocation of terminal pins	Gage
102.	Defective seal around terminal plate	Visual
103.	Glass seal cracked around two squib pins	Visual
104.	Case damaged	Visual
105.	Objective evidence relative to material missing or improper (see 3.1)	Visual
106.	Evidence of poor workmanship (see 3.11)	Visual
Minor:		
201.	Incorrect or illegible marking	Visual
202.	Serialization improper or missing (see 3.4)	Visual

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4.3.2.1.4 Preparation for delivery (see 5).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Preservation, packing material, packaging and marking not in accordance with FED-STD-356	Visual
Minor:		
	201. Evidence of poor workmanship (see 3.11)	Visual

4.3.3 Testing. All tests shall be performed. Tests shall be performed in the order specified below:

Test	Test Classification
a. Contact position and resistance	Major
b. Bridge resistance	Major
c. Insulation resistance	Major
d. X-ray	Major
e. Squib energy, non-function	Major
f. Squib energy, functioning	Major
g. Low temperature	Major
h. High temperature	Major
i. Leakage	Major

Of these tests, f, g and h are considered destructive and samples so tested shall not be returned to the lot. Destructive samples are over and above the deliverable quantity of items required by the contract.

4.3.3.1 Contact position and resistance, bridge resistance, insulation resistance, x-ray, leakage and squib energy, non-function. Each switch shall be tested in accordance with 4.4, applicable paragraph. If any item fails to comply with (3.6.1.1 for Type I switches and 3.6.1.2 for Type II switches), (3.7.1 for Type I switches and 3.7.2 for Type II switches), (3.8.1.1 for Type I switches and 3.8.1.2 for Type II switches), (3.9.6), (3.5.1.1 for Type I switches and 3.5.1.2 for Type II switches), the item shall be rejected and removed from the lot.

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4.3.3.2 Squib energy functioning (see 3.5.2.1 for Type I switches and 3.5.2.2 for Type II switches), low temperature (see 3.9.1) and high temperature (see 3.9.2). After passing first article tests, a quantity of 200 switches shall be produced without any additional testing. After this initial group is produced, a sample of 6 switches shall be selected from each future inspection lot of 100 switches and subjected to the tests specified in Table X. If any switch fails to comply with the applicable requirement, the inspection lot shall be rejected.

Table X

Sequence of Tests	No. of Sample Units Tested	No. of Sample Units Non-Function Tested & Then Functioned
Squib energy, functioning	2	2
Low temperature	4	2
High temperature	2	2

4.3.4 Acceptance inspection equipment.

4.3.4.1 Major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL) (see EL's-8867522 for Type I switches and EL-8871618 for Type II switches). When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment and obtain approval prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired (see 6.3).
b. The code "CD" is listed in the "Part of Identifying Number" column on the EL (see 6.1.2).

4.3.4.2 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on ELs. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

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4.4.1 Squib energy non-function. The switch shall be subjected to the discharge of a capacitor of sufficient capacity to deliver 15,000 ergs min when charged to a voltage of 15 ± 1 volts direct current (vdc). Series dropping resistors may be used to dissipate any energy in excess of 15,000 ergs. The switch shall then be checked to determine whether it had functioned. The bridge wire shall then be subjected to 100 ma dc for 10 seconds. Again, the switch shall be checked to determine whether it had functioned.

4.4.2 Squib energy, functioning. The switch squib connected in series with a resistor of 1.1 ± 1% ohm shall be subjected to the discharge from 57 ± 1 microfarad capacitor charged to 15 vdc maximum. The switch's functioning time, the period from the application of voltage to the squib to the functioning of the switch, shall be measured with suitable equipment. After functioning the switch shall be tested for compliance with the requirements of contact position and resistance, 3.6.2.1 for Type I switches and 3.6.2.2 for Type II switches and insulation resistance, 3.8.2.1 for Type I switches and 3.8.2.2 for Type II switches.

4.4.3 Contact position and resistance. The contact resistance of the switch shall be measured with a suitable resistance measuring instrument to determine which contacts are closed and the resistance of those contacts.

4.4.4 Bridge resistance. The bridge resistance of the switch shall be measured with a Kelvin Bridge Ohmmeter 4286 or equivalent. The maximum current output of the meter shall be 5 ma.

4.4.5 Insulation resistance. The insulation resistance of the switch shall be measured with an insulation resistance tester having an output of 500 ± 50 vdc. The maximum current output of the tester shall be 5 ma. The insulation resistance shall be measured between those terminals specified in 3.8, as applicable.

4.4.6 Environmental testing (see 3.9).

4.4.6.1 Low temperature. The switch shall be placed in a test chamber at ambient humidity. The chamber temperature shall be lowered to -65 ± 5°F and maintained for 4 hours after stabilization (see 6.4). After the 4 hour period and while still at -65°F temperature, the switch shall be tested for compliance with the requirements of squib energy (3.5.1.1 for Type I switches and 3.5.1.2 for Type II switches), contact position (3.6.1.1 for Type I switches and 3.6.1.2 for Type II switches), bridge resistance (3.7.1 for Type I switches and 3.7.2 for Type II switches) and insulation resistance (3.8.1.1 for Type I switches and 3.8.1.2 for Type II switches) followed by squib energy 3.5.2.1 for Type I switches and 3.5.2.2 for Type II switches.

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4.4.6.2 High temperature. The switch shall be placed in a test chamber at ambient humidity. The chamber temperature shall be raised to 160 + 5°F and maintained for 4 hours after stabilization (see 6.4). After the 4 hour period and while still at 160°F temperature, the switch shall be tested for compliance with the requirements of squib energy (3.5.1.1 for Type I switches and 3.5.1.2 for Type II switches), contact position (3.6.1.1 for Type I switches and 3.6.1.2 for Type II switches), bridge resistance (3.7.1 for Type I switches and 3.7.2 for Type II switches) and insulation resistance (3.8.1.1 for Type I switches and 3.8.1.2 for Type II switches) followed by squib energy (3.5.2.1 for Type I switches and 3.5.2.2 for Type II switches).

4.4.6.3 Shock. The switch shall be mounted in a shock device and subjected to a total of 18 shocks of 30 ± 5 g's, three in each direction along each of the three mutually perpendicular axes. The time duration of each shock shall be 11 ± 2 millisec (ms) as measured between points which are 10 percent of peak value. The time to peak shall be 5.5 ± 1 ms. Upon completion of the shocks the switch shall be tested for compliance with the requirements of squib energy (3.5.1.1 for Type I switch and 3.5.1.2 for Type II switches), contact position (3.6.1.1 for Type I switches and 3.6.1.2 for Type II switches), bridge resistance (3.7.1 for Type I switches and 3.7.2 for Type II switches) and insulation resistance (3.8.1.1 for Type I switches and 3.8.1.2 for Type II switches), followed by squib energy (3.5.2.1 for Type I switches and 3.5.2.2 for Type II switches).

4.4.6.4 Transportation vibration. The switch shall be mounted on a vibration machine and vibrated for 15 minutes in each of three mutually perpendicular axes at the frequencies and amplitudes specified in 3.9.4 with one 8 to 500 to 8 Hertz sweep every 15 minutes. The total test time shall be 45 minutes. Upon completion of the vibrations, the switch shall be tested for compliance with the requirements of squib energy (3.5.1.1 for Type I switches and 3.5.1.2 for Type II switches), contact position (3.6.1.1 for Type I switches and 3.6.1.2 for Type II switches), bridge resistance (3.7.1 for Type I switches and 3.7.2 for Type II switches) and insulation resistance (3.8.1.1 for Type I switches and 3.8.1.2 for Type II switches), followed by operational vibrations (see 3.9.5).

4.4.6.5 Operational vibration. The switch shall be mounted on a vibration machine and vibrated in three mutually perpendicular planes, the longitudinal plane being conducted last, using a standard noise or complex wave tape. The tape shall have the characteristics listed in Table XI.

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Table XI

Time (Sec)	Type of Signal	Ratio of Test Level To Calibration Level	RMS g Level
0-20	Noise (Calibration) (1)	1	Set gain to give 2 g RMS
20-30	None	-	-
30-31	Noise (1)	11	22
31-46	Noise (1) plus sinusoid (2)	4.25	8.5
	Noise alone	3.75	7.5
	Sinusoid alone	2.	4.
46-61	Noise (1) plus sinusoid (3)	8.5	17.
	Noise alone	7.5	15.
	Sinusoid alone	4.	8.
61-99	Noise (1) plus sinusoid (4)	4.25	8.5
	Noise alone	3.75	7.5
	Sinusoid alone	2.	4.
99-100	Noise (1)	11.	22.

(1) Band limited (15 - 1500 Hz) white Gaussian Noise.
(2) Sine wave swept from 800 to 1300 Hz at a rate increasing directly with frequency. One full sweep in 15 seconds.
(3) Sine wave swept from 1300 to 800 Hz at a rate decreasing directly with frequency. One full sweep in 15 seconds.
(4) Sine wave swept from 15 to 1500 Hz and back to 15 Hz at a rate increasing directly with frequency. One full cycle in 38 seconds.

For creating the random noise, the output of a General Radio random noise generator, or its equivalent, may be used if proper care is taken to insure the correct amplitude distribution of the signal. For testing the noise generator or any other associated equipment, the use of an amplitude distribution analyzer is suggested. For insuring the correct noise bandwidth, a filter with an asymptotic slope of at lease 24 decibels (Db) per octave and 3 Db points at 15 and 1500 Hz shall be considered acceptable.

4.4.6.5.1 Method and procedure with the standard noise tape. Before applying the tape, the transfer function or frequency response of the entire vibration system shall be as flat as is consistent with the present state of the art. Plus or minus 3 Db or better over the band from 15 to 1500 Hz shall be considered an objective in this respect. After then adjusted to that with the noise calibration signal, the g level listed in 4.4.6.5 is obtained. With the same gain setting, the tape is run for the remainder of the test and the g levels obtained will conform to the levels listed in the Table of 4.4.6.5.

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4.4.6.5.2 During section 4 of 3.9.5, in the longitudinal plane, the switch shall be tested for compliance with the requirements of squib energy (3.5.2.1 for Type I switches and 3.5.2.2 for Type II switches) and after the completion of the vibrations shall be tested for compliance with the requirements of contact position and resistance (3.6.2.1 for Type I switches and 3.6.2.2 for Type II switches) and insulation resistance (3.8.2.1 for Type I switches and 3.8.2.2 for Type II switches).

4.4.6.6 Leakage (applicable to Type II switches only). The switch shall be immersed in 2 inches of water and all trapped air bubbles wiped free. The air over the surface of the water shall be evacuated to a pressure of 5 ± 1 psi absolute, while at this pressure, no bubbles shall appear from or upon the switch during a 60 second period (minimum).

4.4.6.7 Temperature humidity (applicable to Type II switches only). The switch shall be placed within the test chamber and subjected to the humidity test cycle as specified in MIL-E-5272, Humidity tests, Procedure I. All temperatures shall be maintained at ± 5°F. Three such cycles shall be performed consecutively. The ph value shall be adjusted by the use of chemically pure hydroxide. Upon completion of the last cycle, the switch shall be removed from the test chamber and with the moisture removed by wiping, shall be tested for compliance with squib energy (3.5), contact position and resistance (3.6), bridge resistance (3.7 and insulation resistance (3.8).

4.4.7 Radiographic inspection. All switches shall be examined in accordance with MIL-STD-271. The switches shall be X-rayed in two longitudinal positions, with terminal pins up and with terminal pins on the side.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing and marking. Preservation, packaging, packing and marking shall be in accordance with FED-STD-356 or as specified in the contract or purchase order.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

a. Title, number, and date of this specification.

b. Type and "M" number of the switch to be procured.

6.1.2 Contract data requirements (see 4.3.4.1). Data specified in 4.3.4.1 will be delivered as identified on a numbered DD Form 1664 when required on a DD Form 1423 incorporated in the contract.

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6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commander, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SARPA-QA-T.

6.4 Temperature stabilization. The temperature is considered stabilized when the greatest peaks (both highest and lowest) continually observed over 5 minutes min do not exceed the nominal temperature plus tolerance specified.

6.5 Supersession data. This specification supersedes the requirements of MIL-S-48796(PA), dated 27 January 1975, and MIL-S-48549(PA) dated 27 January 1975.

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Preparing activity:
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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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